

[BUG] QNode with expval incorrect output dtype for `diff_method="parametric"

<https://github.com/PennyLaneAI/pennylane/issues/2368>

ROW 21

[BUG] QNode with expval incorrect output dtype for diff_method="parameter-shift" #2368

Closed

#2367



antalszava opened on Mar 22, 2022 · edited by antalszava

Edits ...

Expected behavior

The `dtype` specified on the device is considered for the QNode output and single-precision can be set.

Actual behavior

Certain cases always yield double-precision.

Additional information

The importance of the mismatch comes into play when TensorFlow and Torch with other tools may produce errors for outputs with double-precision.

Source code

```
import tensorflow as tf
import numpy as np
import pennylane as qml

nwires = 5

dev = qml.device("lightning.qubit", wires=nwires, shots=None)
dev.R_DTYPE = np.float32
dev.C_DTYPE = np.complex64

@qml.qnode(dev, interface="tf")
def circuit():
    return qml.expval(qml.PauliZ(0))

circuit().dtype
```

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.23.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /pennylane
Requires: numpy, scipy, networkx, retworkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, pe
Required-by: PennyLane-Orquestra, pennylane-qulacs, PennyLane-Honeywell, PennyLane-AQT, PennyLane-PQ, PennyLar
Platform info: Linux-5.13.0-35-generic-x86_64-with-glibc2.10
Python version: 3.8.5
Numpy version: 1.22.3
Scipy version: 1.7.3
Installed devices:
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
```

Microsoft Outl

Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.

Home-page: <https://github.com/XanaduAI/pennylane>

Author:

Author-email:

License: Apache License 2.0

Location: /pennylane

Requires: numpy, scipy, networkx, networkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, pe

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- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.22.0)
- cirq.pasqal (PennyLane-Cirq-0.22.0)
- cirq.qsim (PennyLane-Cirq-0.22.0)
- cirq.qsimh (PennyLane-Cirq-0.22.0)
- cirq.simulator (PennyLane-Cirq-0.22.0)
- qiskit.aer (PennyLane-qiskit-0.22.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq.sampler (PennyLane-qiskit-0.22.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.5.7.dev0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.5.7.dev0)
- strawberryfields.fock (PennyLane-SF-0.21.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.21.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.21.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.21.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.21.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.23.0.dev2)
- default.gaussian (PennyLane-0.23.0.dev0)
- default.mixed (PennyLane-0.23.0.dev0)
- default.qubit (PennyLane-0.23.0.dev0)
- default.qubit.autograd (PennyLane-0.23.0.dev0)
- default.qubit.jax (PennyLane-0.23.0.dev0)
- default.qubit.tf (PennyLane-0.23.0.dev0)
- default.qubit.torch (PennyLane-0.23.0.dev0)

Existing GitHub issues

- ☒ I have searched existing GitHub issues to make sure the issue does not already exist.



antalszava added bug on Mar 22, 2022



antalszava mentioned this on Mar 22, 2022

[Fix QNode incorrect dtype output for `qml.expval` / `qml.var` / `single` `qml.probs` \(not backprop\) #2367](#)



antalszava changed the title BUG QNode incorrect output dtype BUG QNode with expval incorrect output dtype for ``diff_method="parameter-shift"`` on Mar 24, 2022

Fix QNode incorrect dtype output for `qml.expval/qml.var/single qml.probs` (not backprop) #2367

<> Code

Merged rmoyard merged 21 commits into `master` from `fix_incorrect_dtype` on Mar 25, 2022

Conversation 4 Commits 21 Checks 0 Files changed 10 +209 -11



antalszava commented on Mar 22, 2022 • edited

Contributor

Context:

Computing the expectation value with a non-backprop device doesn't preserve the device specific dtype. See example in the linked issue.

Note that this issue is more general, though the fix is specific. For more failing cases (including backprop case) see issue #2374.

Description of the Change:

Updates the logic in `QubitDevice` to preserve the user-defined dtype in the internal computation.

Benefits:

Computing the expectation value, variance and statevector in a QNode with `diff_method="parameter-shift"` preserves the types defined by the user on the device. If `qml.probs` is the only measurement, this also applies then.

Possible Drawbacks:

The internal `R_DTYPE` and `C_DTYPE` objects may have to be defined explicitly if a special object is outputted when computing expectation values, variances, probabilities or quantum states. See the `SpecialObservable` related test case changes.

Related GitHub Issues:

Closes #2368



antalszava added 2 commits 3 years ago



update the dtypes for backprop

e5084b7



convert results to array

094bbc2



github-actions bot commented on Mar 22, 2022

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Reviewers

rmoyard



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[BUG] QNode with expval incorrect output dt...

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